

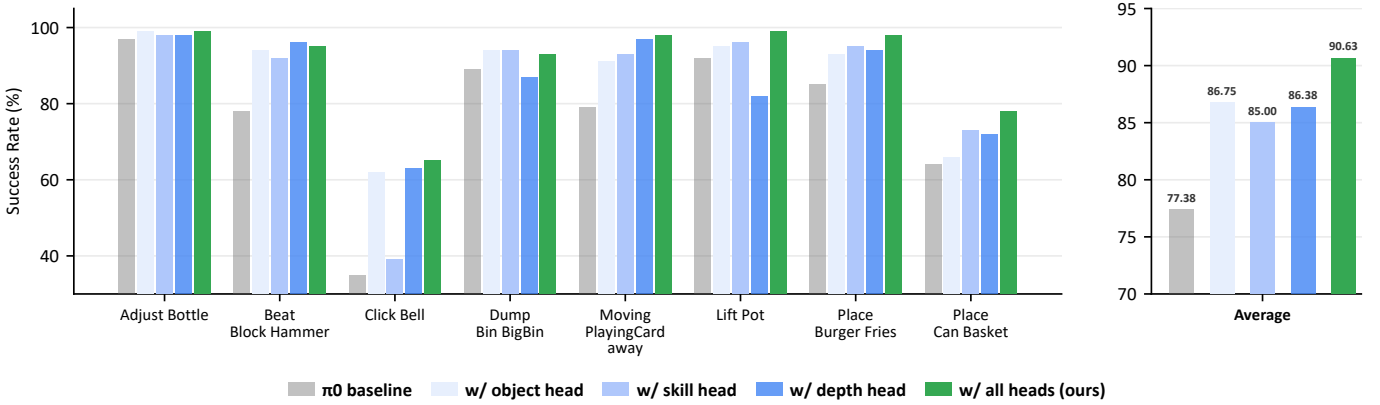


Fig. 5: **RoboTwin 2.0 Benchmark Performance.** Success rates across 8 manipulation tasks comparing the π_0 baseline, single-head experts, and our full model. While specific heads excel at aligned tasks (e.g., depth head for geometry-heavy Beat Hammer Block), the full model (purple) integrates these capabilities to achieve the best overall average performance (90.63%).

two bowls and place on rack, and (3) *clean the tabletop*: sweep trash with broom/dustpan, pour into tray. On PSI-Bot RealMan, we design three chemistry-lab manipulation tasks: (4) *place beaker in heating mantle*: grasp a beaker and insert it into the heating mantle, (5) *stack beakers*: nest small beakers inside a large one, and (6) *heat beaker*: place an asbestos mesh on the iron stand and then place the beaker on the mesh. These laboratory tasks focus on the manipulation challenges posed by transparent, rigid objects and tight geometric constraints. They do not evaluate complete safety-critical chemical procedures; in particular, *heat beaker* requires placing the beaker onto the heating setup rather than controlling the heating process itself. Detailed success criteria are provided in Appendix L2.

Evaluation protocol: For each task and model, we perform 20 trials. A trial is successful if the entire task is completed.

Generalization evaluation: We evaluate three generalization settings on both platforms: *in-domain generalization*, *scene generalization*, and *lighting generalization*. Here, *in-domain generalization* focuses on object position variations within the training distribution, while preserving task semantics and scene layout. *Scene generalization* introduces distracting objects into the workspace, testing robustness to clutter and semantic interference. *Lighting generalization* varies illumination intensity and color temperature, assessing sensitivity to perceptual shifts. Results are summarized in Table II, with detailed setting definitions in Appendix M.

V. ANALYSIS

In this section, we aim to answer the following questions:

- 1) Do VLAs under-utilize vision-language representations in action decoding process, and can explicit factor guidance close this gap? (Section V-B)
- 2) Does our proposed GuidedVLA improve baseline performance under both in-distribution and out-of-distribution evaluations? (Section V-A)
- 3) Which factors (object, skill, geometry) matter most for which task types? (Section V-A)

- 4) Does our attention head specialization indeed lead to learning decoupled features? (Section V-C)
- 5) How different architectural choices for guidance influence performance? (Section V-E)

A. Task-suite Analysis and Cross-benchmark Generalization

We analyze how each factor contributes to different task suites and use representative results from simulation and real-world evaluations to explain *why* each specified head helps.

Object Head: Visual Generalization. Tasks involving clutter or distractors necessitate a precise understanding of object instance identities. On the LIBERO-Plus *Object* suite, which stresses object-level distinctions, the object head yields the strongest single-head result (82.5%, +8.4% over π_0 , Table I; full results in Appendix B). This aligns with the intuition that explicit object-centric representations mitigate grounding failures, allowing the policy to filter out irrelevant visual cues that confuse the baseline.

Skill Head: Temporal Coherence. Long-horizon manipulation requires maintaining “stage awareness” to transition correctly between sub-skills. On LIBERO-Plus, the skill head gives the best single-head result on the *Goal* suite (68.9%) and remains above π_0 on the *Long* suite (62.7% vs. 60.1%, Table I). Similarly, for the *Lift Pot* task (RoboTwin 2.0) involving a strict sequence of grasping, stabilizing, and lifting, this head achieves the best single-head success rate (96%, Fig. 5; full results in Appendix B). These results validate that explicit skill recognition provides the temporal scaffolding needed to prevent premature termination or mode collapse during phase transitions.

Depth Head: Geometric Precision. Tasks reliant on precise 3D localization, such as pressing or insertion, necessitate accurate depth estimation which 2D backbone alone often fails to provide. On RoboTwin 2.0, the *Click Bell* task requires precise Z-axis control to trigger the mechanism without collision; the depth head drastically improves performance from 35% to 63% (Fig. 5). We observe a similar trend in *Beat Hammer*